

CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education
Advanced Subsidiary Level and Advanced Level

MATHEMATICS

9709/04

Paper 4 Mechanics 1 (M1)

May/June 2003

1 hour 15 minutes

Additional materials: Answer Booklet/Paper
Graph paper
List of Formulae (MF9)

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.
Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

Where a numerical value for the acceleration due to gravity is needed, use 10 m s^{-2} .

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 50.

Questions carrying smaller numbers of marks are printed earlier in the paper, and questions carrying larger numbers of marks later in the paper.

The use of an electronic calculator is expected, where appropriate.

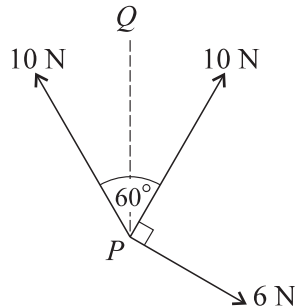
You are reminded of the need for clear presentation in your answers.

This document consists of 4 printed pages.



- 1 A crate of mass 800 kg is lifted vertically, at constant speed, by the cable of a crane. Find
- (i) the tension in the cable, [1]
 - (ii) the power applied to the crate in increasing the height by 20 m in 50 s. [3]

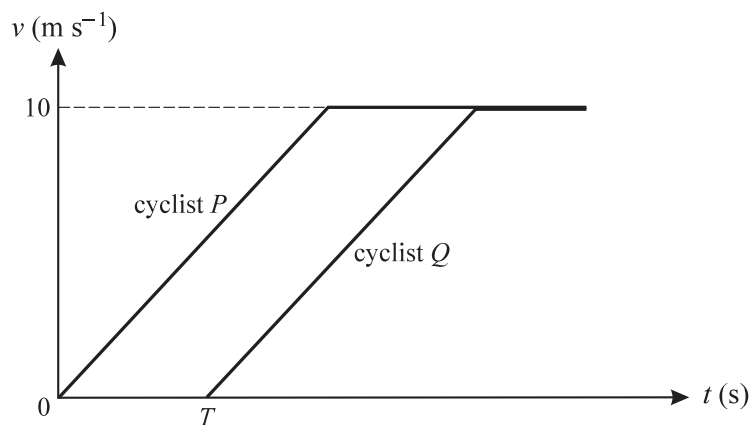
2



Three coplanar forces of magnitudes 10 N, 10 N and 6 N act at a point P in the directions shown in the diagram. PQ is the bisector of the angle between the two forces of magnitude 10 N.

- (i) Find the component of the resultant of the three forces
 - (a) in the direction of PQ , [2]
 - (b) in the direction perpendicular to PQ . [1]
- (ii) Find the magnitude of the resultant of the three forces. [2]

3



The diagram shows the velocity-time graphs for the motion of two cyclists P and Q , who travel in the same direction along a straight path. Both cyclists start from rest at the same point O and both accelerate at 2 m s^{-2} up to a speed of 10 m s^{-1} . Both then continue at a constant speed of 10 m s^{-1} . Q starts his journey T seconds after P .

- (i) Show in a sketch of the diagram the region whose area represents the displacement of P , from O , at the instant when Q starts. [1]

Given that P has travelled 16 m at the instant when Q starts, find

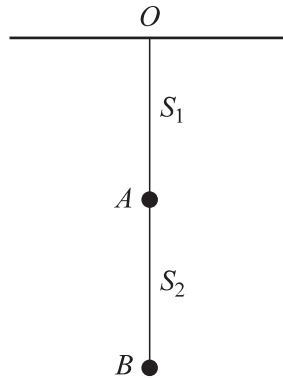
- (ii) the value of T , [3]
- (iii) the distance between P and Q when Q 's speed reaches 10 m s^{-1} . [2]

- 4 A particle moves in a straight line. Its displacement t seconds after leaving the fixed point O is x metres, where $x = \frac{1}{2}t^2 + \frac{1}{30}t^3$. Find

(i) the speed of the particle when $t = 10$, [3]

(ii) the value of t for which the acceleration of the particle is twice its initial acceleration. [3]

5



S_1 and S_2 are light inextensible strings, and A and B are particles each of mass 0.2 kg. Particle A is suspended from a fixed point O by the string S_1 , and particle B is suspended from A by the string S_2 . The particles hang in equilibrium as shown in the diagram.

(i) Find the tensions in S_1 and S_2 . [3]

The string S_1 is cut and the particles fall. The air resistance acting on A is 0.4 N and the air resistance acting on B is 0.2 N.

(ii) Find the acceleration of the particles and the tension in S_2 . [5]

- 6 A small block of mass 0.15 kg moves on a horizontal surface. The coefficient of friction between the block and the surface is 0.025 .

(i) Find the frictional force acting on the block. [2]

(ii) Show that the deceleration of the block is 0.25 m s^{-2} . [2]

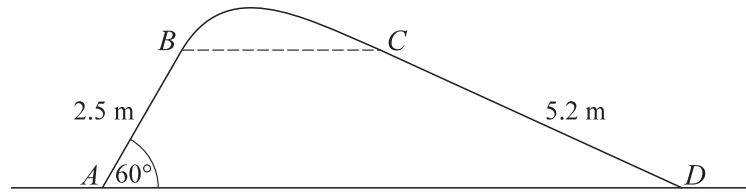
The block is struck from a point A on the surface and, 4 s later, it hits a boundary board at a point B . The initial speed of the block is 5.5 m s^{-1} .

(iii) Find the distance AB . [2]

The block rebounds from the board with a speed of 3.5 m s^{-1} and moves along the line BA . Find

(iv) the speed with which the block passes through A , [2]

(v) the total distance moved by the block, from the instant when it was struck at A until the instant when it comes to rest. [2]



The diagram shows a vertical cross-section $ABCD$ of a surface. The parts AB and CD are straight and have lengths 2.5 m and 5.2 m respectively. AD is horizontal, and AB is inclined at 60° to the horizontal. The points B and C are at the same height above AD . The parts of the surface containing AB and BC are smooth. A particle P is given a velocity of 8 m s^{-1} at A , in the direction AB , and it subsequently reaches D . The particle does not lose contact with the surface during this motion.

- (i) Find the speed of P at B . [4]
- (ii) Show that the maximum height of the cross-section, above AD , is less than 3.2 m. [2]
- (iii) State briefly why P 's speed at C is the same as its speed at B . [1]
- (iv) The frictional force acting on the particle as it travels from C to D is 1.4 N. Given that the mass of P is 0.4 kg, find the speed with which P reaches D . [4]